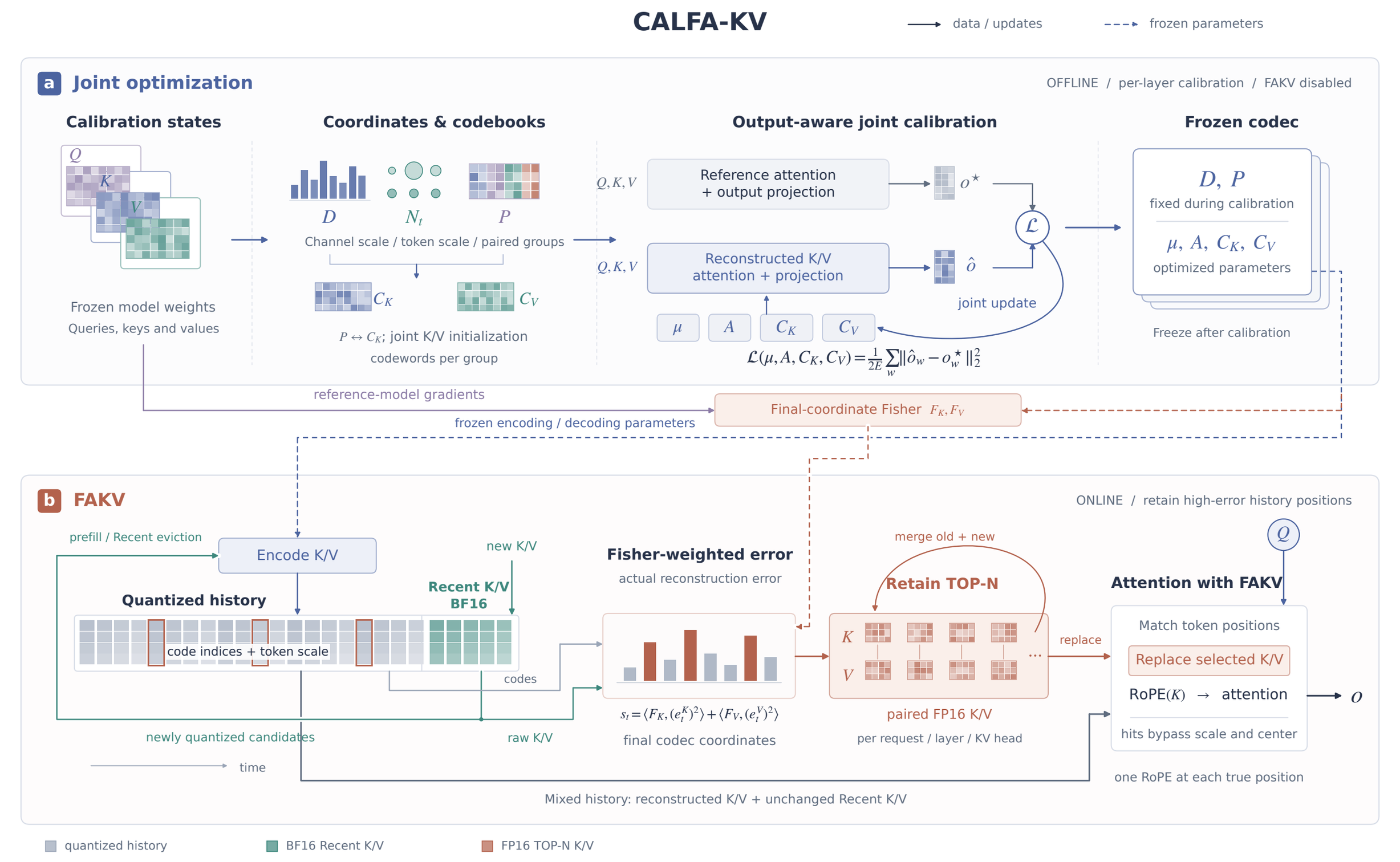

CALFA-KV
data / updates
frozen parameters
a Joint optimization
OFFLINE / per-layer calibration / FAKV disabled
Calibration states
Coordinates & codebooks
Output-aware joint calibration
Frozen codec
Q
K
V
Frozen model weights
Queries, keys and values
D
N_t
P
Channel scale / token scale / paired groups
C_K
C_V
P ↔ C_K; joint K/V initialization
codewords per group
Q,K,V
Reference attention
+ output projection
o*
Q,K,V
Reconstructed K/V
attention + projection
ô
joint update
μ
A
C_K
C_V
L(μ, A, C_K, C_V) = 1/2E Σ_w ||ô_w − o*_w||_2^2
D, P
fixed during calibration
μ, A, C_K, C_V
optimized parameters
Freeze after calibration
reference-model gradients
frozen encoding / decoding parameters
Final-coordinate Fisher F_K, F_V
b FAKV
ONLINE / retain high-error history positions
prefill / Recent eviction
Encode K/V
new K/V
Recent K/V
BF16
Quantized history
code indices + token scale
codes
Fisher-weighted error
actual reconstruction error
s_t = ⟨F_K, (e_t^K)^2⟩ + ⟨F_V, (e_t^V)^2⟩
final codec coordinates
merge old + new
Retain TOP-N
K
V
paired FP16 K/V
per request / layer / KV head
replace
Q
Attention with FAKV
Match token positions
Replace selected K/V
RoPE(K) → attention
O
hits bypass scale and center
one RoPE at each true position
newly quantized candidates
raw K/V
time
Mixed history: reconstructed K/V + unchanged Recent K/V
quantized history
BF16 Recent K/V
FP16 TOP-N K/V